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From IGC records to measurable trajectories

Cleaning decisions and their unresolved effects

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Chapter 2 • 20-minute supervisor meeting

The archive contains recorded cross-country flights

	Paragliders	Hang gliders
Catalogue entries	203,343	9,259
Downloaded IGC logs	186,052	6,716
Retained flights	156,305	6,093
Retention	84.0%	90.7%

FFVL Coupe Fédérale de Distance: self-declared flights, heterogeneous recorders, sites and seasons.

The analysis population is defined jointly by who records and declares a flight, and by which records our criteria retain.

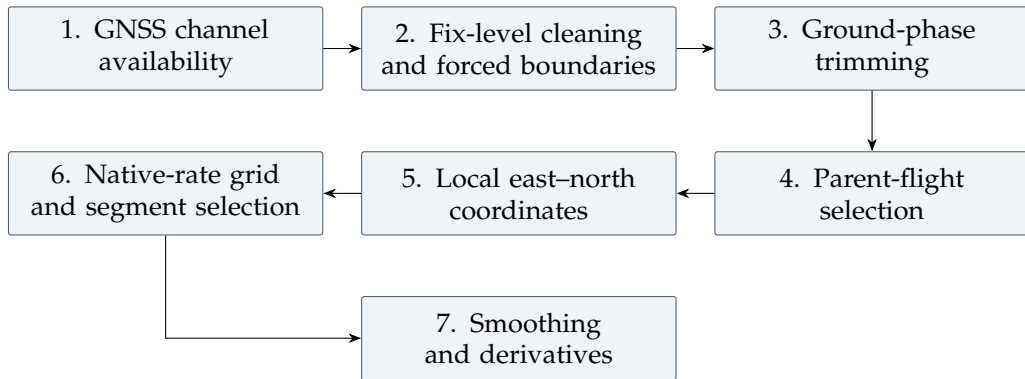
Current pipeline census. Para catalogue seasons: 1999–2000 to 2025–2026; hang: 2001–2002 to 2025–2026. Downloaded-track coverage differs.

Each decoded fix supplies

Time and horizontal position; GNSS and pressure altitude when recorded; receiver validity declaration.

Neither the catalogue nor the retained sample is an unbiased population of all soaring flights.

The processing order determines what each cut acts on



Parent-flight restrictions: recorded duration ≥ 40 min, path ≥ 20 km, altitude range ≥ 600 m, elapsed span ≤ 16 h, plus integrity and reach checks.

These restrictions select a study population. Genuine short flights or nearly level soaring can fail them. Flight-level cuts are not reapplied to every later segment.

Adopted altitude channel

One channel throughout the population:
GNSS presence $\geq 95\%$ and range ≥ 30 m at flight level.

Missing or invalid GNSS altitude becomes missing z ; horizontal coordinates remain subject to their own tests.

The barometer does not substitute for GNSS altitude.

Keep usable horizontal trajectories; assess witness-dependent results with a paired-coverage sensitivity analysis.

Locally available corroboration

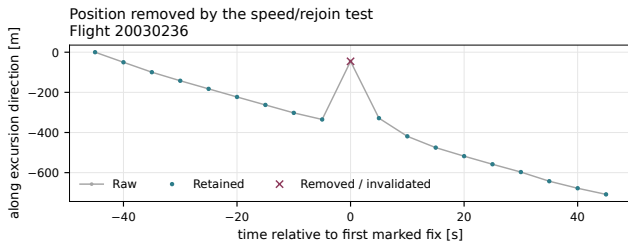
$$B_I = B_{\text{flight}} \prod_{i \in I} b_i.$$

Here $b_i = 1$ for a finite, nonzero pressure reading at fix i , and 0 otherwise.

The candidate interval needs simultaneous position and pressure readings. An incomplete local witness abstains.

Eligible barometer: flight presence $\geq 95\%$, range ≥ 30 m. Eligibility occurs in 72.1% para / 84.8% hang flights attempted by the pipeline.

A Hampel flag describes a departure; the reach/rejoin rule removes it



Horizontal removal requires

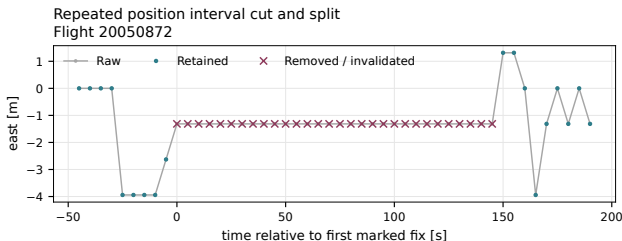
1. An unreachable position from the last accepted fix.
2. A reachable rejoin after removing the block.

$$v_{xy}^{\max} = 45 \text{ m/s (para)}, 55 \text{ m/s (hang)}.$$

Resolved: the Hampel flag is not required for deletion. A long corrupt block can contaminate the median and evade that flag.

Real paraglider flight 20030236. The displayed coordinate is a projection used to show the defect; the rule uses geodetic horizontal distances. [Identifier formula and assumptions](#).

A frozen-position candidate is removed and split



Three joint conditions:

$$D_I < 15 \text{ m}, \quad T_I \geq 60 \text{ s}, \quad W_I = 1.$$

D_I is the horizontal bounding-box diagonal.

The witness uses exact repetition, locally complete barometry, or a stated recorder-declaration fallback.

A removed frozen candidate never becomes an interpolated straight flight. Its unobserved motion stays separated.

Real paraglider flight 20050872. These signatures support a cleaning action; they do not prove the receiver's internal lock state. [Complete witness branches](#).

A corrected decision rule



One missing pressure fix:
the barometric branch abstains.

Frozen-run counts changed in 188 cases; interior-ground counts in 2, with overlap possible. These counts measure algorithmic impact, not detector accuracy.

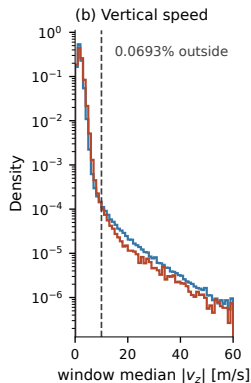
Targeted old/new-rule comparison

2,388 candidates; 189 flights changed.

22,468 additional horizontal fixes retained; no flight admission changed in that targeted set.

The complete archive was subsequently rebuilt with the corrected rule.

The vertical-speed threshold is now 10 m/s



The threshold lies near the slower-decaying tail and the crossing of the two discipline densities around 9–10 m/s.

Above 10 m/s: 0.069% of window medians in the diagnostic sample.

This is an exceedance fraction, not the fraction of altitudes removed. Both neighbouring decisions still enter the rule.

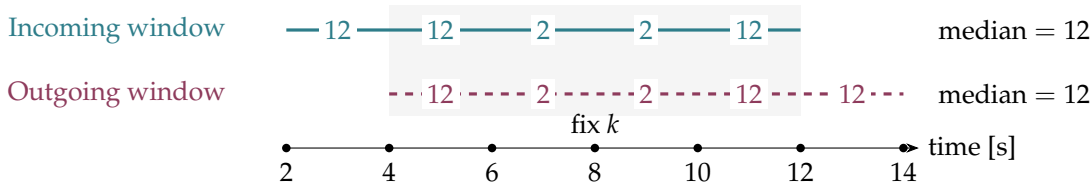
Does changing the threshold materially alter vertical features or phase statistics?

The curve motivates an operational cutoff. It does not establish a physical speed maximum or prove that every larger value is an outlier.

Two window medians determine one altitude decision

With at least five finite steps in a window,

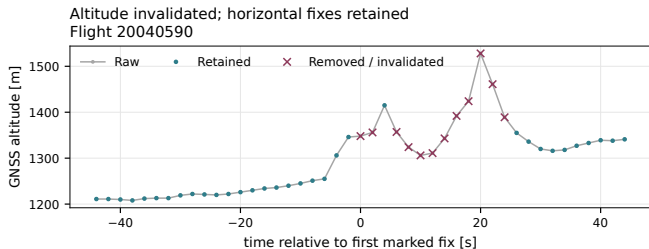
$$u_j = \text{median}_{|m_i - m_j| \leq 5 \text{ s}} |v_{z,i}^{\text{obs}}|, \quad m_i = (t_i + t_{i+1})/2; \quad z_k \rightarrow \text{missing if } u_{k-1} > 10 \wedge u_k > 10.$$



Both medians exceed the threshold although the two steps touching fix k are only 2 m/s. This is a neighbourhood decision.

Schematic with 2-s cadence; speeds in m/s. Below five finite steps, $u_j = |v_{z,j}^{\text{obs}}|$ instead of the median. Equality at 10 m/s does not trigger invalidation.

Altitude invalidation preserves the horizontal trajectory



Crosses mark altitudes invalidated by the current implementation.

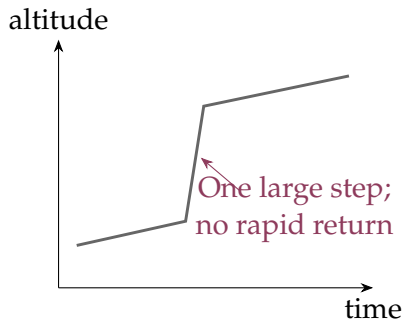
Horizontal fixes remain, unless they fail their own tests.

Reconstructed altitude is identified later by a separate quality flag.

An inspectable decision is useful evidence of implementation behavior. It does not supply an independent label of the true altitude.

Real paraglider flight 20040590. [Synthetic spike and sustained-excess examples](#) isolate what the separate tests do.

An unreturned altitude step remains an unresolved case



A single jump may leave the median low and fail the out-and-back test.

Candidate counters occur in [47,341 para / 1,349 hang](#) retained-flight records, before later trimming and segment selection.

A targeted check found candidate times inside usable phase-feature windows in all eight inspected examples.

Compare exclusion or splitting near candidates before deciding whether an automatic correction is justified.

Schematic. Candidate counts do not count confirmed sensor defects surviving in the final trajectories. No automatic level-shift correction is implemented.



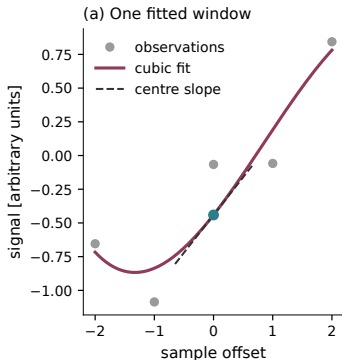
30-s gap ● — ● — ● — ● — ● ● — ● — ● Split

Missing z  Reconstruct z only

Current limitation: altitude-only reconstruction has no duration cap. Quality masks, rather than the time-gap rule, protect vertical analyses.

Examples at $\Delta t = 5$ s, hence $g_{\max} = 20$ s. Existing cleaning boundaries always split. Segment duration ≥ 90 s; interpolated grid fraction $< 10\%$.

One local polynomial estimates position and derivatives



For offsets $u = -m, \dots, m$,

$$P_i(u) = \sum_{k=0}^3 a_k(i) u^k,$$

$$\hat{x}_i = a_0,$$

$$\hat{v}_i = a_1 / \Delta t,$$

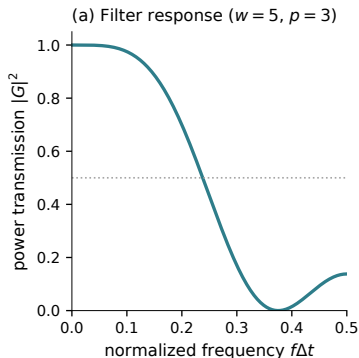
$$\hat{a}_i = 2a_2 / \Delta t^2.$$

The same fit is applied separately to E, N, z inside each segment.

Smoothing controls derivative noise, but it also changes resolved motion and propagates reconstructed values through its windows.

Synthetic illustration of the implemented Savitzky–Golay fit. [Savitzky & Golay \(1964\)](#).

The nominal 5-s timescale does not mean a 5-s window for every logger



$$w = \max\{\lceil 5 \text{ s} / \Delta t \rceil_{\text{odd}}, 5\}.$$

Five samples span:

Cadence	Window span
1 s	4 s
3 s	12 s
5 s	20 s

The filter's frequency response depends on cadence.

Quantify the sensitivity of 10-s transport and phase features to cadence and the smoothing window.

The observed spectral knee motivates a working timescale; it is not the exact cutoff of the Savitzky–Golay filter. The response shown is the interior value-filter response.

Most rejected paraglider records fail to yield a usable segment

First recorded rejection reason	Para	Hang
No segment survives resampling	20,899	194
Altitude range below 600 m	4,899	155
Recorded duration below 40 min	2,084	79
All other rejection reasons	1,865	195
Retained	156,305	6,093

The largest loss makes sampling and segment selection a priority for sensitivity checks.

How do retention and the resulting observables change across recorder cadence, season and region?

Sequential pipeline census. Rejection reasons are mutually exclusive first failures; changing their order would change the attribution.
[Complete census.](#)

What has been checked, and what still needs empirical validation?

Completed

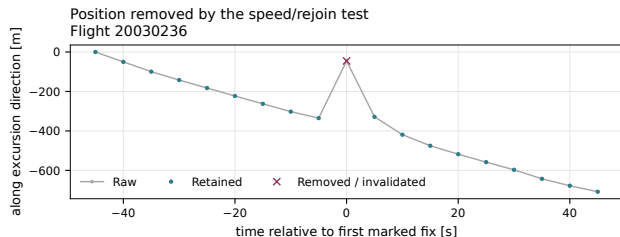
- Current cleaning and all 38 rebuild stages.
- Rule tests, reconstruction masks and trajectory examples.
- Reproduction of stored cleaned outputs for 250 retained flights per discipline.
- Reconciliation of numerical reports and manuscript definitions.

Still open

- Independent detector error rates.
- Observable sensitivity to thresholds, smoothing and cadence.
- Candidate altitude steps and retained suspect intervals.
- GNSS height-datum consistency where it affects conclusions.

Prioritize checks that could change the transport conclusions or the phase features.

Numerical reproducibility and mathematical consistency are necessary. They do not establish physical correctness of every retained fix or every selection rule.



Open the Python viewer

Paraglider flight **20030236**
Position-spike example around 4410
s after the raw record begins.

Compare the raw and current
processed trajectory, then zoom into
the suspect interval.

Local macOS link; the companion launcher must be registered. The viewer can open without the SSD; this raw flight needs the connected archive. Offline fallback: the exact frozen example above.

Which cleaning uncertainties should we resolve first?

1. Vertical reliability.

Compare observables with and without neighbourhoods of candidate altitude steps; assess an explicit limit on altitude-only reconstruction.

2. Sensitivity of the scientific results.

Vary the 10-m/s threshold and smoothing/cadence choices on a fixed comparison population; report changes in transport and phase features.

3. Independent evidence for difficult decisions.

Review paired-channel examples and realistic injected defects; distinguish acceptance criteria from measured error rates.

The rebuilt trajectories are reproducible. Their remaining uncertainties must be quantified where they affect the physics.

Discussion material

Extra figures, derivations and references

The identifier's scale is a convention, not a false-alarm probability

$$r_k = d((\varphi_k, \lambda_k), (\tilde{\varphi}_k, \tilde{\lambda}_k)), \quad \sigma_k = 1.4826 \operatorname{median}_{|t_j - t_k| \leq 20 \text{ s}} r_j.$$

Flag when $r_k > \max(5\sigma_k, 15 \text{ m})$, with at least five fixes in the window.

- Component-wise medians define the local reference position; residuals use horizontal great-circle distance.
- The factor 1.4826 is the one-dimensional Gaussian MAD normalization. For these radial residuals it is an operational scale convention.
- The absolute reach/rejoin gate acts independently of this diagnostic flag.

Identifier terminology: [Davies & Gather \(1993\)](#), Example 2.2. Thresholds and GPS removal rules are choices of this analysis.

The complete frozen-position witness

Let E_I mean exact coordinate repetition; d_I is the fraction with invalid GNSS declaration or missing GNSS altitude. Then

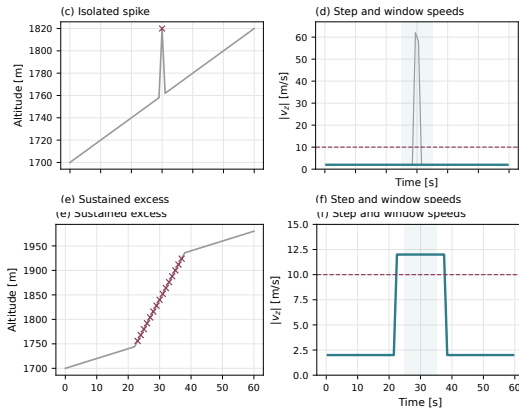
$$W_I = E_I \vee \begin{cases} |\Delta b_I| < 5 \text{ m}, & B_I = 1, \\ \text{false}, & B_{\text{flight}} = 1, B_I = 0, \\ d_I > 1/2, & B_{\text{flight}} = 0. \end{cases}$$

Δb_I is the last-quarter minus first-quarter pressure-altitude median.

For interior ground detection, the pressure test instead uses a detrended 5–95% span and a slope guard, with sustained low speed for at least 10 min. Missing local pressure always causes that guard to abstain.

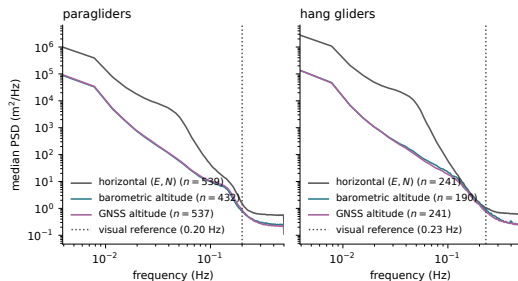
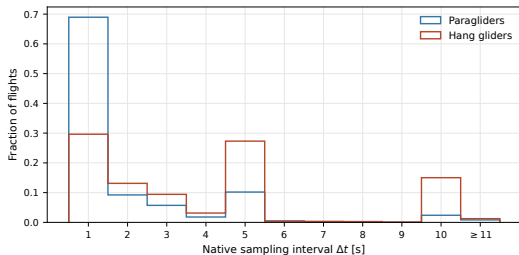
Small net pressure change can hide an excursion; distinct sensors do not guarantee independent recorder errors. Shorter qualifying slow intervals are flagged and retained.

An isolated spike and a sustained excess trigger different evidence



Synthetic data processed by the actual implementation. The spike scarcely changes a populated-window median; excessive opposite-sign bounding steps trigger its separate test. The sustained excess raises both local medians.

Cadence and the spectral motivation for smoothing



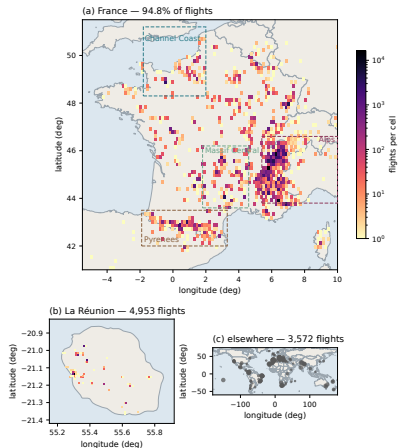
The raw spectral knee around 0.2 Hz motivates a working timescale. The recorder population and the Savitzky–Golay response still require explicit cadence controls.

The complete first-failure census

Why the flight was dropped	paragliders	hang gliders
unreadable track	27	0
no usable GNSS altitude	697	135
never airborne	197	1
integrity gate	550	53
shorter than the duration cut	2,084	79
longer than the duration cut	3	0
path below the floor	173	1
altitude activity below the floor	4,899	155
out of reach of its own first fix	217	5
no segment survived resampling	20,899	194
no segment carries the smoothing window	1	0
retained	156,305 (84.0%)	6,093 (90.7%)

Current clean archive, pipeline version 2.2.0. All numerical entries are imported from the frozen generated table.

Geographic coverage is part of the sampling problem



Launch sites and flight characteristics are unevenly represented.

Equipment, season, recording technology and environment can change together.

Directional transport is analysed in Chapter 3; this chapter establishes which observations reach that analysis.

Savitzky & Golay (1964)

Polynomial smoothing and derivatives

Fritsch & Butland (1984)

Monotone piecewise cubic interpolation

Davies & Gather (1993)

Hampel identifier

Numerical sources: current Chapter 2 census, witness comparison and real cleaning examples. Figure and value snapshots are identified by SHA-256 hashes in the companion manifest.

The cited methods do not establish the validity of the chosen numerical thresholds, the recorder status of individual fixes or the accuracy of downstream phase labels.